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St.MOTHER THERESA ENGINEERING COLLEGE

COMPUTER SCIENCE ENGINEERING

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REG NO: 953023104099

Completed the project named as

Phase 4

FRONT END TECHNOLOGY

IBE PORTFOLIO WEBSITE

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Phase 4 — Enhancements & Deployment (Week 9)

1. Additional Features

Enhance your portfolio by adding interactive or advanced features.

Examples:

Contact form with email service (EmailJS / Firebase).

Downloadable resume button.

Project filtering (by technology or category).

Theme toggle (light/dark mode).

Testimonials or achievements section.

Implementation:

```
// Example: Theme toggle
const [darkMode, setDarkMode] = useState(false);
return (
  <div className={darkMode ? "dark" : "light"}>
    <button onClick={() => setDarkMode(!darkMode)}>Toggle Theme</button>
  </div>
);
```

2. UI/UX Improvements

Focus on making the site look professional and user-friendly.

Tasks:

Use consistent color palette and typography.

Add smooth animations (Framer Motion or AOS).

Ensure responsiveness across devices using media queries or Tailwind CSS.

Add hover effects for buttons and project cards.

Tools: Tailwind CSS, Framer Motion, React Icons.

Example:

<motion.div whileHover={{ scale: 1.05 }}>

<button className="bg-blue-600 text-white px-4 py-2 rounded-lg">View Project</button>

</motion.div>

3. API Enhancements

If your portfolio pulls data dynamically (e.g., GitHub projects, blogs, or contact form), optimize APIs.

Steps:

Integrate GitHub API to show your repositories.

Use Axios or Fetch API for calling data.

Handle loading and error states gracefully.

Example:

```
useEffect(() => {

fetch("https://api.github.com/users/<username>/repos")
    .then(res => res.json())
    .then(data => setRepos(data))
```

```
    .catch(err => console.log(err));
}, []);
```

4. Performance & Security Checks

Before deployment, ensure your app is fast and secure.

Checklist:

Run Lighthouse Audit (in Chrome DevTools).

Optimize images (WebP format, lazy loading).

Use HTTPS and environment variables for API keys.

Avoid console logs and unused imports.

5. Testing of Enhancements

Ensure everything works correctly before final deployment.

Actions:

Test across browsers (Chrome, Edge, Safari).

Check responsiveness on mobile/tablet.

Verify navigation, forms, and links.

Use libraries like Jest or React Testing Library for component testing.

6. Deployment (Netlify / Vercel / Cloud Platform)

Finally, deploy your project live for public access.

Netlify Deployment:

Push code to GitHub.

Go to Netlify.

Click New Site → Import from GitHub.

Select your repo and deploy.

Vercel Deployment:

Go to Vercel.

Import your GitHub repo.

Set build command: npm run build.

Your site goes live with a free custom URL.

Additional Features	Add
resume, contact form, dark mode	React,
EmailJS	

UI/UX Improvements		
		Enhance look, animations, responsiveness
		Tailwind, Framer Motion
API Enhancements	Fetch dynamic data	Axios, GitHub API
Performance & Security		
		Optimize speed and safety
		Lighthouse, HTTPS
Testing	Verify new changes	Jest, Manual
Deployment	Host live project	
		Netlify / Vercel

Would you like me to combine Phase 3 + Phase 4 into a single detailed implementation document

with sample code sections for your Naan Mudhalvan IBM Portfolio submission?